

Linux System Administration: File Management, Permissions, and User/Group Configuration

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Part A: File & Directory Management Tasks

1. Create three files:
file1.txt, file2.txt, file3.txt

```
(dolon@NMDolon)-[~]  
$ touch file1.txt file2.txt file3.txt  
  
(dolon@NMDolon)-[~]  
$ ls  
Assignment  Downloads  file3.txt  PGD-08  Task  
Desktop     file1.txt  'Lab Practice'  Pictures  Templates  
Documents   file2.txt  Music        Public    Videos
```

2. Create three directories:
directory1, directory2, directory3

```
(dolon@NMDolon)-[~]  
$ mkdir directory1 directory2 directory3  
  
(dolon@NMDolon)-[~]  
$ ls  
Assignment  directory2  Downloads  file3.txt  PGD-08  Task  
Desktop     directory3  file1.txt  'Lab Practice'  Pictures  Templates  
directory1  Documents  file2.txt  Music        Public    Videos
```

3. Copy file1.txt into directory1/

```
(dolon@NMDolon)-[~]  
$ cp file1.txt directory1  
  
(dolon@NMDolon)-[~]  
$ ls directory1  
file1.txt
```

4. Move file2.txt into directory2/

```
(dolon@NMDolon)-[~]  
$ mv file2.txt directory2  
  
(dolon@NMDolon)-[~]  
$ ls directory2  
file2.txt
```

5. Rename file3.txt to renamed_file5.txt

```
(dolon@NMDolon)-[~]
$ mv file3.txt renamed_file5.txt

(dolon@NMDolon)-[~]
$ ls
Assignment  directory3  ' Lab Practice'  Public  Videos
Desktop    Documents  Music            renamed_file5.txt
directory1  Downloads  PGD-08          Task
directory2  file1.txt  Pictures        Templates
```

6. Delete file1.txt from the original location

```
(dolon@NMDolon)-[~]
$ rm file1.txt

(dolon@NMDolon)-[~]
$ ls
Assignment  directory3  Music  renamed_file5.txt
Desktop    Documents  PGD-08  Task
directory1  Downloads  Pictures  Templates
directory2  ' Lab Practice'  Public  Videos
```

7. List all files and folders using:

- o Ls

```
(dolon@NMDolon)-[~]
$ ls
Assignment  directory3  Music  renamed_file5.txt
Desktop    Documents  PGD-08  Task
directory1  Downloads  Pictures  Templates
directory2  ' Lab Practice'  Public  Videos
```

- ls -l
- ls -a

```
(dolon@NMDolon)-[~]
$ ls -l
total 60
drwxrwxr-x 5 dolon dolon 4096 Nov 27 08:23 Assignment
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:18 Desktop
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:37 directory1
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:37 directory2
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:36 directory3
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Documents
drwxr-xr-x 2 dolon dolon 4096 Nov 25 03:04 Downloads
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 ' Lab Practice'
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Music
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 PGD-08
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:09 Pictures
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Public
-rw-rw-r-- 1 dolon dolon 0 Nov 27 08:36 renamed_file5.txt
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:20 Task
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Templates
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Videos

(dolon@NMDolon)-[~]
$ ls -a
. Pictures
.. .profile
Assignment Public
.bash_logout renamed_file5.txt
.bashrc .ssh
.bashrc.original .sudo_as_admin_successful
.cache Task
.config Templates
Desktop .vboxclient-clipboard-tty7-control.pid
directory1 .vboxclient-clipboard-tty7-service.pid
directory2 .vboxclient-display-svga-x11-tty7-control.pid
directory3 .vboxclient-draganddrop-tty7-control.pid
.dnrc .vboxclient-draganddrop-tty7-service.pid
Documents .vboxclient-hostversion-tty7-control.pid
Downloads .vboxclient-seamless-tty7-control.pid
.face .vboxclient-seamless-tty7-service.pid
.face.icon .vboxclient-vmsvga-session-tty7-control.pid
.gnupg .vboxclient-vmsvga-session-tty7-service.pid
.ICEauthority Videos
.java .wget-hsts
' Lab Practice' .Xauthority
.local .xsession-errors
.mozilla .xsession-errors.old
.msf4 .zprofile
Music .zsh_history
PGD-08 .zshrc
```

8. View the contents of any file using:

- cat
- more

```
(dolon@NMDolon)~  
$ echo "This is a assignment task by Niyaz Mursid Dolon" > renamed_file5.txt
```

```
(dolon@NMDolon)-[~]
$ cat renamed_file5.txt
This is a assignment task by Niyaz Mursid Dolon
```

```
(dolon@NMDolon)-[~]  
$ more renamed_file5.txt  
This is a assignment task by Niyaz Mursid Dolon
```

less

```
(dolon@NMDolon)-[~]  
$ less renamed_file5.txt
```

[illegible]

Part B: Compression & Decompression

1. Compress directory1 using the following tools:
 - zip
 - tar

```
(dolon@NMDolon)-[~]
$ zip directory1.zip -r directory1
adding: directory1/ (stored 0%)
adding: directory1/file1.txt (stored 0%)

(dolon@NMDolon)-[~]
$ tar -cvf directory1.tar directory1
directory1/
directory1/file1.txt

(dolon@NMDolon)-[~]
$ ls
Assignment      directory2      Music           Task
Desktop         directory3      PGD-08         Templates
directory1      Documents      Pictures       Videos
directory1.tar  Downloads      Public
directory1.zip  'Lab Practice' renamed_file5.txt
```

2. Compress file1.txt using the following tools:
 - gzip
 - bzip2

```
(dolon@NMDolon)-[~]
$ cp directory1/file1.txt .

(dolon@NMDolon)-[~]
$ gzip file1.txt

(dolon@NMDolon)-[~]
$ cp directory1/file1.txt .

(dolon@NMDolon)-[~]
$ bzip2 file1.txt

(dolon@NMDolon)-[~]
$ ls
Assignment      directory2      file1.txt.gz    Public
Desktop         directory3      'Lab Practice'  renamed_file5.txt
directory1      Documents      Music           Task
directory1.tar  Downloads      PGD-08         Templates
directory1.zip  file1.txt.bz2  Pictures       Videos

(dolon@NMDolon)-[~]
$
```

3. Decompress/extract above number 1 & 2 each and verify the files:

- unzip
- tar -xvf
- gunzip

```
(dolon@NMDolon)-[~]
$ unzip directory1.zip
Archive:  directory1.zip
replace directory1/file1.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: yes
extracting: directory1/file1.txt

(dolon@NMDolon)-[~]
$ tar -xvf directory1.tar
directory1/
directory1/file1.txt

(dolon@NMDolon)-[~]
$ gunzip file1.txt.gz

(dolon@NMDolon)-[~]
$ ls
Assignment      directory2      file1.txt.bz2   Public
Desktop         directory3      ' Lab Practice' renamed_file5.txt
directory1       Documents      Music           Task
directory1.tar  Downloads      PGD-08         Templates
directory1.zip  file1.txt      Pictures        Videos
```

- bunzip2

```
(dolon@NMDolon)-[~]
$ rm file1.txt

(dolon@NMDolon)-[~]
$ ls
Assignment      directory2      ' Lab Practice' renamed_file5.txt
Desktop         directory3      Music           Task
directory1       Documents      PGD-08         Templates
directory1.tar  Downloads      Pictures        Videos
directory1.zip  file1.txt.bz2  Public

(dolon@NMDolon)-[~]
$ bunzip2 file1.txt.bz2

(dolon@NMDolon)-[~]
$ ls
Assignment      directory2      ' Lab Practice' renamed_file5.txt
Desktop         directory3      Music           Task
directory1       Documents      PGD-08         Templates
directory1.tar  Downloads      Pictures        Videos
directory1.zip  file1.txt      Public
```

Part C: File & Folder Permissions

1. Assign **octal permissions**:
 - file1.txt: chmod 644 file1.txt
 - directory1: chmod 755 directory1

```
(dolon@NMDolon)-[~]
$ ls -l
total 80
drwxrwxr-x 5 dolon dolon 4096 Nov 27 08:23 Assignment
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:18 Desktop
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:37 directory1
-rw-rw-r-- 1 dolon dolon 10240 Nov 27 14:45 directory1.tar
-rw-rw-r-- 1 dolon dolon 340 Nov 27 14:44 directory1.zip
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:37 directory2
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:36 directory3
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Documents
drwxr-xr-x 2 dolon dolon 4096 Nov 25 03:04 Downloads
-rw-rw-r-- 1 dolon dolon 0 Nov 27 15:02 file1.txt
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 'Lab Practice'
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Music
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 PGD-08
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:09 Pictures
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Public
-rw-rw-r-- 1 dolon dolon 48 Nov 27 08:39 renamed_file5.txt
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:20 Task
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Templates
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Videos

(dolon@NMDolon)-[~]
$ chmod 644 file1.txt

(dolon@NMDolon)-[~]
$ chmod 755 directory1

(dolon@NMDolon)-[~]
$ chmod 700 directory2/file2.txt

(dolon@NMDolon)-[~]
$ ls -l
total 80
drwxrwxr-x 5 dolon dolon 4096 Nov 27 08:23 Assignment
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:18 Desktop
drwxr-xr-x 2 dolon dolon 4096 Nov 27 08:37 directory1
-rw-rw-r-- 1 dolon dolon 10240 Nov 27 14:45 directory1.tar
-rw-rw-r-- 1 dolon dolon 340 Nov 27 14:44 directory1.zip
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:37 directory2
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:36 directory3
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Documents
drwxr-xr-x 2 dolon dolon 4096 Nov 25 03:04 Downloads
-rw-r--r-- 1 dolon dolon 0 Nov 27 15:02 file1.txt
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 'Lab Practice'
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Music
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 PGD-08
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:09 Pictures
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Public
-rw-rw-r-- 1 dolon dolon 48 Nov 27 08:39 renamed_file5.txt
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:20 Task
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Templates
```


file2.txt: chmod 700 file2.txt

```
(dolon@NMDolon)-[~]
$ ls -l directory2/file2.txt
-rwx----- 1 dolon dolon 0 Nov 27 08:36 directory2/file2.txt
```

2. Assign **symbolic permissions**:

- Remove all permissions from file2.txt
- Add read/write to owner, read to group to file2.txt
- Add execute permission to others to file2.txt

```
(dolon@NMDolon)-[~]
$ chmod 000 directory2/file2.txt

(dolon@NMDolon)-[~]
$ ls -l directory2/file2.txt
----- 1 dolon dolon 0 Nov 27 08:36 directory2/file2.txt

(dolon@NMDolon)-[~]
$ chmod u+rw,g+r directory2/file2.txt

(dolon@NMDolon)-[~]
$ chmod o+x directory2/file2.txt

(dolon@NMDolon)-[~]
$ ls -l directory2/file2.txt
-rw-r--x 1 dolon dolon 0 Nov 27 08:36 directory2/file2.txt
```

3. Show permission changes using ls -l

```
(dolon@NMDolon)-[~]
$ ls -l
total 80
drwxrwxr-x 5 dolon dolon 4096 Nov 27 08:23 Assignment
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:18 Desktop
drwxr-xr-x 2 dolon dolon 4096 Nov 27 08:37 directory1
-rw-rw-r-- 1 dolon dolon 10240 Nov 27 14:45 directory1.tar
-rw-rw-r-- 1 dolon dolon 340 Nov 27 14:44 directory1.zip
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:37 directory2
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:36 directory3
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Documents
drwxr-xr-x 2 dolon dolon 4096 Nov 25 03:04 Downloads
-rw-r--r-- 1 dolon dolon 0 Nov 27 15:02 file1.txt
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 'Lab Practice'
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Music
drwxrwxr-x 2 dolon dolon 4096 Nov 25 03:13 PGD-08
drwxr-xr-x 2 dolon dolon 4096 Nov 27 00:09 Pictures
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Public
-rw-rw-r-- 1 dolon dolon 48 Nov 27 08:39 renamed_file5.txt
drwxrwxr-x 2 dolon dolon 4096 Nov 27 08:20 Task
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Templates
drwxr-xr-x 2 dolon dolon 4096 Nov 22 07:54 Videos

(dolon@NMDolon)-[~]
$ ls -l directory2/file2.txt
-rw-r--x 1 dolon dolon 0 Nov 27 08:36 directory2/file2.txt
```

Part D: User and Group Management

1. Create three users:

user1, user2, user3

→ Command: `sudo adduser user1` (and so on)

```
(dolon@NMDolon)-[~]
$ sudo adduser user1
[sudo] password for dolon:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user1
Enter the new value, or press ENTER for the default
    Full Name []: Niyaz
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] yes

(dolon@NMDolon)-[~]
$ sudo adduser user2
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
    Full Name []: Mursid
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] yes

(dolon@NMDolon)-[~]
$ sudo adduser user3
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user3
Enter the new value, or press ENTER for the default
    Full Name []: Dln
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] yes
```

```

(dolon@NMDolon)-[~]
$ cat /etc/passwd | grep user
sshd:x:988:65534:sshd user:/run/sshd:/usr/sbin/nologin
cups-pk-helper:x:119:121:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
pipewire:x:986:986:system user for pipewire:/nonexistent:/usr/sbin/nologin
user1:x:1001:1001:Niyaz,,,:/home/user1:/bin/bash
user2:x:1002:1002:Mursid,,,:/home/user2:/bin/bash
user3:x:1003:1003:Dln,,,:/home/user3:/bin/bash

```

2. Create three groups:
group1, group2, group3
→ Command: `sudo groupadd group1`

```

(dolon@NMDolon)-[~]
$ sudo groupadd group1

(dolon@NMDolon)-[~]
$ sudo groupadd group2

(dolon@NMDolon)-[~]
$ sudo groupadd group3

```

```

(dolon@NMDolon)-[~]
$ cat /etc/group | grep group
nogroup:x:65534:
group1:x:1004:
group2:x:1005:
group3:x:1006:

```

3. Add users to groups:

- Add user1 to group1
- Add user2 (existing) to group1
- Add a **new user** user4 to **existing group** group2
- Add a **new user** user5 and add them to a **new group** group4

```
(dolon@NMDolon)-[~]  
$ sudo usermod -aG group1 user1  
[sudo] password for dolon:  
  
(dolon@NMDolon)-[~]  
$ sudo usermod -a -G group1 user2  
  
(dolon@NMDolon)-[~]  
$ sudo useradd -G group2 user4  
  
(dolon@NMDolon)-[~]  
$ sudo groupadd group4 && sudo useradd -m -G group4 user5 &&  
sudo passwd user5  
New password:  
Retype new password:  
passwd: password updated successfully
```

```
dolon:x:1000:1000:Niyaz Mursid Dolon,,,:/home/dolon:/usr/bin/zsh  
pcscd:x:984:984:PC/SC Smart Card Daemon:/:usr/sbin/nologin  
user1:x:1001:1001:Niyaz,,,:/home/user1:/bin/bash  
user2:x:1002:1002:Mursid,,,:/home/user2:/bin/bash  
user3:x:1003:1003:Dln,,,:/home/user3:/bin/bash  
user4:x:1004:1007::/home/user4:/bin/sh  
user5:x:1005:1009::/home/user5:/bin/sh
```

```
dolon:x:1000:  
vboxsf:x:132:dolon  
kaboxer:x:133:dolon  
pcscd:x:984:  
user1:x:1001:  
user2:x:1002:  
user3:x:1003:  
group1:x:1004:user1,user2  
group2:x:1005:user4  
group3:x:1006:  
user4:x:1007:  
group4:x:1008:user5  
user5:x:1009:
```